Q. Find the following Limits

1. $\displaystyle \underset{(x,y)\to(0,\pi/4)}{Lt} \sec x \tan y$

2. $\displaystyle \underset{(x,y)\to(0,0)}{Lt} \frac{2+y}{x+\cos y}$

Q Show that the limits of the following functions do not exist as $(x,y)\to(0,0)$

1. $\displaystyle f(x,y) = \frac{x+y}{x-y}$

2. $\displaystyle f(x,y) = \frac{y}{x^2-y}$

Q Show that $f(x,y) = x^2+2y^2$ is Continuous at $(1,2)$

Q Discuss the Continuity of the function

$$f(x,y) = \begin{cases} 3xy & \text{if } (x,y)\neq(1,2) \\ 0 & \text{if } (x,y)=(1,2) \end{cases}$$

as $(x,y)\to(1,2)$

Q If $w = \cos(x+y) + \cos(x-y)$

Show that $\dfrac{\partial^2 w}{\partial x^2} = \dfrac{\partial^2 w}{\partial y^2}$

Q Find all second order Partial derivatives of the following functions

$f(x,y) = \sin xy$, $f(x,y) = \tan^{-1}(y/x)$, $f(x,y) = \sec xy$

Q Find f_x, f_y, f_z, f_{xy}, f_{yz}, and f_{zx} for the following functions

i) $f(x,y,z) = xy + yz + zx$

ii) $f(x,y,z) = yz\ln(xy)$

iii) $f(x,y,z) = z\sin^{-1}(y/x)$

iv) $f(x,y,z) = e^{-(x^2+y^2+z^2)}$

v) $f(x,y,z) = x^2y^3\sin z + e^{2x}$

Q If $f(x,y,z) = \sin xyz$ then find f_{xyz}

Q If $f(x,y) = \cos x\sinh y + \sin x\cosh y$ then show that

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

Q If $f(x,y,z) = 2z^2 - 3(x^2+y^2)z$, then show that

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \neq 0$$

Q Find $\frac{dy}{dx}$ for the following function

i) $f(x,y) = y^2 + xy - 12$

ii) $f(x,y) = xy - e^x\sin y$

iii) $f(x,y) = \ln(x^2+y^2) - \tan^{-1}(y/x)$

Q Find the linearization of $f(x,y) = x^2 - xy + \frac{1}{2}y^2 + 3$ at the point $(3,2)$

Q Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ for the following functions

Using the Chain Rule

i) $w = \ln(x^2+y^2+z^2)$, $x = re^s \sin r$, $y = re^s \cos r$,
$z = re^s$

ii $w = x^2+y^2+z^2$, $x = r^2 e^s$, $y = r \tan s$, $z = \frac{\ln r}{s}$

Q If $f(x,y) = \ln\sqrt{x^2+y^2}$
$f(x,y) = e^{-2x}\cos 2y + e^{-2y}\cos 2x$
then Show that $f_{xx} + f_{yy} = 0$

Q: Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for the following functions

$f(x,y) = e^{xy}\ln y$, $f(x,y) = \frac{1}{2}\ln(x^2+y^2) + \sin^{-1}\left(\frac{y}{2x}\right)$

$f(x,y) = \frac{1}{2}e^{x^2+y^2} + \tan^{-1}(x/y)$, $f(x,y) = \cosh(x^2+y^2)$

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